

Single-Cell Atlas of AML Reveals Age-Related Gene Regulatory Networks in t(8;21) AML

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eLife Assessment

This manuscript provides a single-cell transcriptomic atlas for AML (222 samples comprising 748,679 cells) integrating data from multiple studies. They use this dataset to investigate t(8;21) AML, and they reconstruct the Gene Regulatory Network and enhancer Gene Regulatory Network, which allowed identification of interesting targets. This aggregation is **useful** and can help infer differences in genetic regulatory modules based on the age of disease onset, which may help explain age-related variations in prognosis and disease development. However, result interpretations and the motivation and critical analysis of the applied computational methods are **incomplete**, and the statistical analyses lack control experiments and should be improved to avoid potential selection bias in the later analyses.

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Abstract

Background

Acute myeloid leukemia (AML) is characterized by cellular and genetic heterogeneity, which correlates with clinical course. Although single-cell RNA sequencing (scRNA-seq) reflect this diversity to some extent, the low sample numbers in individual studies limit the analytic potential of comparisons of specific patient groups.

Results

We performed large scale integration of published scRNA-seq datasets to create a unique single-cell transcriptomic atlas for AML (AML scAtlas), totaling 748,679 cells, from 159 AML patients and 44 healthy donors from 20 different studies. This is the largest single-cell data resource for AML to our knowledge, publicly available at <https://cellxgene.bmh.manchester.ac.uk/AML/>. This AML scAtlas, allowed exploration of the clinical importance of age in t(8;21) AML to an unprecedented level, given the *in-utero* origin of pediatric disease. We uncovered age-associated gene regulatory network (GRN) signatures, which we validated using bulk RNA sequencing data to delineate distinct groups with divergent biological characteristics. Furthermore, using an additional multiomic dataset (scRNA-seq and scATAC-seq), we created a de-noised GRN reflecting the previously defined age-related signatures.

Conclusions

Applying integrated data analysis of the AML scAtlas, we reveal age-dependent gene regulation in t(8;21), perhaps reflecting immature/fetal HSC origin in prenatal origin disease vs postnatal origin. Our analysis revealed that BCLAF1, which is particularly enriched in t(8;21) pediatric AML of inferred *in-utero* origin, is a promising prognostic indicator. The AML scAtlas provides a powerful resource to investigate molecular mechanisms underlying different AML subtypes.

Introduction

Acute myeloid leukemia (AML) is an aggressive blood cancer driven by non-random genomic rearrangements in hematopoietic stem/progenitor cells (HSPCs). Recurrent AML-associated genomic aberrations, which often involve transcriptional or epigenetic regulators, give rise to distinct patterns of gene expression strongly associated with clinical course and chemotherapy response [1, 2]. Single-cell RNA sequencing (scRNA-seq) studies have demonstrated that HSPCs acquire lineage priming at an early stage when still phenotypically immature and disperse down an erythromyeloid or lymphomyeloid differentiation trajectory [3]. In the context of AML, diverse clonal hierarchies include the co-existence of normal hematopoietic clones. Leukemic clones can partially recapitulate myeloid differentiation and have been shown to display functional differences even when defined by the same genotype [4–7]. Indeed, analysis of AML using scRNA-seq has revealed key clonal hierarchies, defining subtype-associated cell types, and dynamic changes following therapy, and have been critical in characterizing leukemic stem cells (LSCs), which propagate the disease and drive relapse [5, 8].

Most AML scRNA-seq studies are limited by small sample numbers and include a mixture of different AML subtypes which may not be directly comparable to one another. Therefore, it is difficult to make biological conclusions with sufficient robustness to be clinically translatable in these individual datasets. To overcome this, we performed large-scale integration of public scRNA-seq datasets to create a single-cell transcriptomic atlas for AML (AML scAtlas). Due to the range of data sources spanning time, geographical locations, and experimental designs, complex batch effects often arise between scRNA-seq datasets which requires a tailored data integration approach [9, 10]. Thus, we benchmarked some widely used batch correction tools [11–13] for our specific data use case.

To begin interrogating the data in the AML scAtlas and demonstrate its utility, given the vast representation of different age groups, we set out to address a developmental question in the biology of AML. The molecular landscape of AML in children is different from adults [2, 14–16]. Correspondingly, the clinical course is different with substantially better outcomes in children. The observed differences in molecular profiles in adult and pediatric AML may, in part, be attributed to the developmental origins of the disease. Acquired chromosomal changes in pediatric leukemia are acquired *in utero*, as leukemia-specific genomic aberrations have been detected in the Gunthrie spots of children who subsequently developed leukemia, sometimes several years later in life [17]. Adult leukemia, in contrast, is believed to develop later in life through acquisition of pre-leukaemic changes and clonal evolution of adult HSPCs [18, 19]. The impact of these developmental stages on leukemia biology are not fully understood, and there is currently no means of quantifying and characterizing differences in the origin of leukemia. However, as childhood AML with presumed *in-utero* origin has a better outcome, for teenagers and young adults, determination of the pre- or postnatal origin might be important for better treatment stratification and prognostication.

AML with t(8;21) (AML-ETO/RUNX1-RUNX1T1) is one of the most frequent AML subtypes in young people, although it affects all ages [2]. The prenatal origins of the t(8;21) rearrangement, has been confirmed even in older children presenting with AML [17]. The prognosis of AML with t(8;21) is better in children than in teenagers and even more so than in young adults [20]. This outcome difference is not fully explainable by co-morbidities and may be related to the developmental origins of the disease. In the intermediate teenage group, t(8;21) AML may comprise both late childhood and early adult disease entities, which may have prognostic implications and could help to explain disease biology and clinical course.

We used our AML scAtlas data resource to characterize age and developmental stage specific signatures in t(8;21) AML by applying single-cell gene regulatory network (GRN) inference methods [21, 22], as a means of revealing cell state heterogeneity across age groups. We then validate and refine our findings in a larger cohort using bulk RNA sequencing (RNA-seq) data from the TARGET [2] and BeatAML [23] studies, defining age-associated GRN signatures and key regulators of t(8;21) AML, which may reflect the developmental origins of the leukemia.

Profiling together both gene expression and chromatin accessibility can be used to decipher the enhancer-driven GRN (eGRN) and enriched transcriptional regulators. Significant heterogeneity across different patients and time points [24] was recently described by analyzing combined scRNA-seq and single-cell Assay for Transposase Accessible Chromatin sequencing (scATAC-seq). This also identified longitudinal transcriptional changes across key clinical milestones, revealing significant heterogeneity across different patients and timepoints. We used the t(8;21) AML data from this study to apply cutting edge GRN inference methods [25], which encompasses both modalities to provide a cleaner GRN which we could correlate with our age-related signatures.

Results

Large Scale Data Integration to Construct a Single-Cell Transcriptomic Atlas of AML (AML scAtlas)

To create AML scAtlas, we combined published scRNA-seq data of primary AML bone marrow samples, from 16 suitable high-quality studies (see Materials and Methods), which included 159 AML samples (Supplementary Table 1). Where on-treatment time points were available, we selected only diagnostic samples to form a reference atlas of primary AML at diagnosis. If studies had healthy donor bone marrow samples these were included, in addition to data from healthy bone marrow samples from four scRNA-seq studies (Supplementary Table 1) to enable comparisons with healthy bone marrow populations. After cell filtering and quality control the

AML scAtlas contains data from 748,679 high quality cells from a total of 20 different scRNA-seq studies [\[4, 6, 8, 26–40\]](#) (Supplementary Figure 1A [\[41\]](#)). Each sample was allocated to an AML clinical subtype, based on the recent European Leukemia Net (ELN) clinical guidelines [\[41\]](#), and a corresponding prognostic risk group. This resource includes a range of molecular subtypes of AML and spans different age groups, including both pediatric and adult AML cases (Figure 1A; 1B [\[41\]](#)). Overall, this dataset is the largest data resource to date for exploring AML biology at single-cell resolution.

From initial analysis of the combined dataset, batch effects were noted, as study-specific clusters were formed. Even within samples of the same study, sample-wise clustering was noted (Supplementary Figure 1A–1C [\[41\]](#)). To address this problem, we performed a benchmark of some widely used batch correction methods (Supplementary Table 2 [\[41\]](#); Supplementary Figure 2A; 2C [\[41\]](#)). This identified scVI as the best method for this dataset (Supplementary Table 2 [\[41\]](#)), which we employed to correct for batch effects. Next, we performed clustering and cell type annotation of AML scAtlas, based on manually curating automated cell type annotation tools results and cluster-wise marker gene expression (Figure 1C; 1D [\[41\]](#)). Cell type proportions analyses across the clinically relevant subtypes in the dataset (Supplementary Figure 3A [\[41\]](#)) show that the AML subtypes were significantly biased towards myeloid cell types (CMP, MEP, GMP, ProMono, CD14+ Mono, CD16+ Mono, cDC, Erythroid) and had a clear predominant cell type, in keeping with AML clonal expansion. On the other hand, healthy donor samples had more balanced cell type proportions and lymphoid cells (T, B, NK, ProB, pDC, Plasma) were well represented (Supplementary Figure 3A [\[41\]](#)).

Due to the established critical role of HSPCs and LSCs in propagating AML and their importance as therapeutic targets [\[42\]](#), we selected HSPC clusters for further analysis, as defined by CD34 expression (Figure 1E [\[41\]](#); Supplementary Figure 3B; 3C [\[41\]](#)). To identify LSCs, we used a curated reference profile of leukaemic stem and progenitor cells (LSPCs) [\[7\]](#) (Figure 1E [\[41\]](#)) and correlated this with calculated LSC6 [\[43\]](#) and LSC17 [\[44\]](#) scores for each cell (Supplementary Figure 3D [\[41\]](#)). We compared the proportions of HSPC/LSPCs across different AML subtypes and risk groups, as defined by the ELN clinical guidelines [\[41\]](#) (Figure 1F [\[41\]](#)). This shows that higher-risk subtypes have a higher proportion of LSCs compared to favorable risk disease.

Application of AML scAtlas to Identifying Age-Associated Gene Regulatory Networks in t(8;21)

The AML scAtlas enables robust comparison of adult and pediatric AML. We hypothesized that in adolescents and young adults with t(8;21)AML, the potential for either *in-utero* or postnatal HSPC origin disease might affect disease biology and prognosis. Thus, we sought to explore biological differences between pediatric and adult cases of t(8;21) AML, aiming to explain and potentially improve prognostication in adolescents and young adults. We selected samples with t(8;21) AML from the AML scAtlas. This included 105,663 cells from 13 adult cases (aged 20–67), 7 adolescent cases (aged 12–17), and 3 pediatric cases (aged 6–8) (Figure 2A–2C [\[41\]](#)). Where gender was not available, this was inferred from ChrY/XIST gene expression (Supplementary Figure 4A [\[41\]](#)). Several adult samples underwent CD34 selection in original studies, excluding more differentiated cell types in these samples. Thus, these cell types were excluded from comparative analysis, focusing only on HSPCs (Figure 2B [\[41\]](#)).

We reconstructed the GRN for the t(8;21) subset of our AML scAtlas using the pySCENIC [\[22\]](#) pipeline and defined critical regulons for this AML subtype. To compare pediatric and adult t(8;21) AML, the top 20 regulons for each of these groups were selected based on the regulon specificity score (RSS) (Supplementary Figure 4B [\[41\]](#)). Unsupervised clustering was performed on the Z score normalized regulon activity score matrix, which highlighted clear differences in the GRN across different age groups (Supplementary Figure 4C [\[41\]](#)). We presume that the differences in the GRN might reflect differences in the pre- or postnatal developmental origins of the disease.

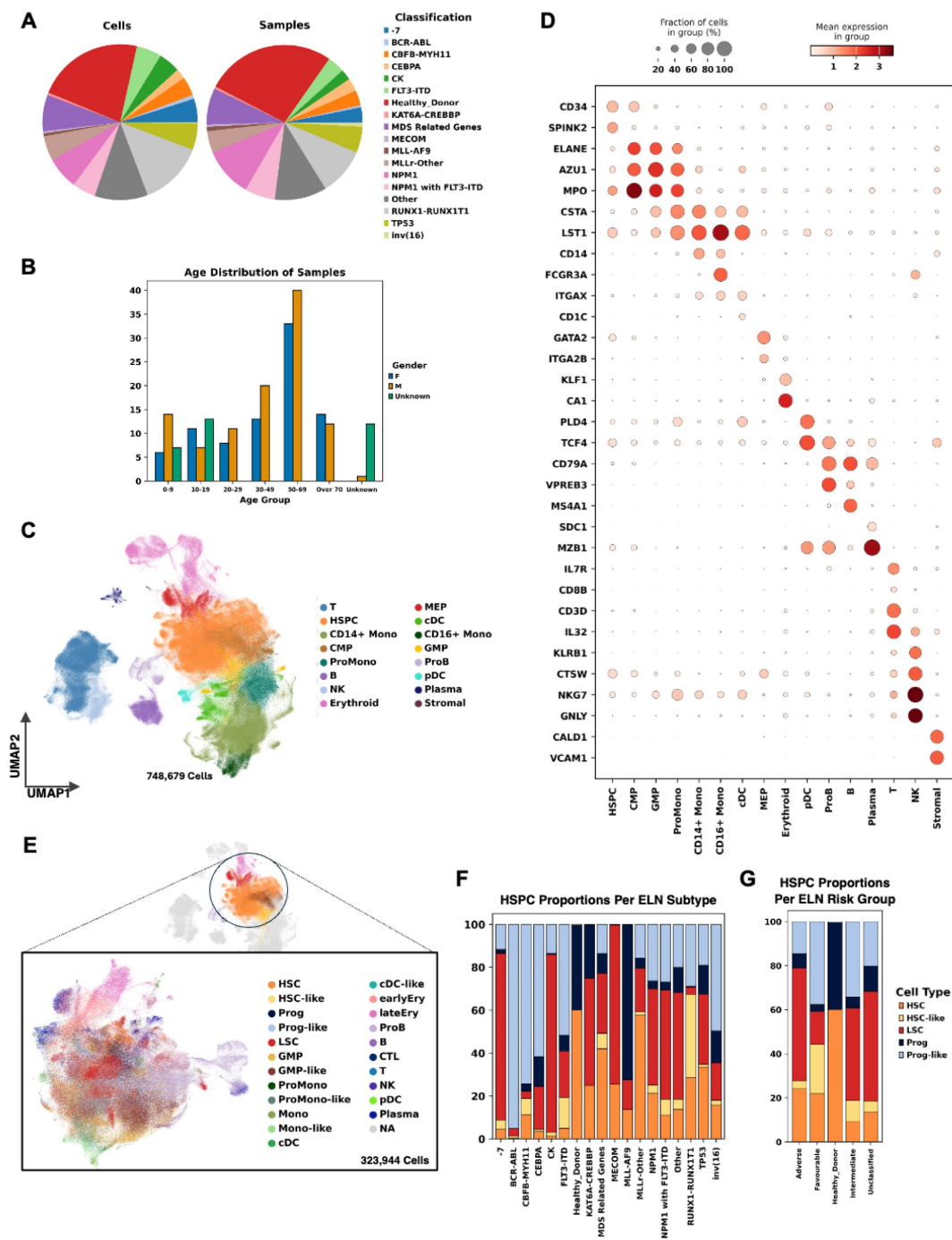


Figure 1 –

Large Scale Data Integration Creates an Atlas of AML

(A) Proportion of cells (left panel) and samples (right panel) belonging to each AML subtype as defined by the ELN clinical guideline. (B) Age group and gender distribution of AML scAtlas cohort samples. (C) scVI harmonized UMAP colored by annotated cell types. (D) The expression of key hematopoietic marker genes across annotated cell types shown on a dotplot. Color scale shows mean gene expression, dot size represents the fraction of cells expressing the given gene. (E) Selected hematopoietic stem and progenitor cell (HSPC) clusters from the AML scAtlas, annotated with a curated reference of leukemia stem and progenitor cells (LSPCs). (F) Proportions of HSPC/LSC populations in each AML subtype as defined by the ELN clinical guideline. (G) Proportions of HSPC/LSC populations in each AML risk group as defined by the ELN clinical guideline.

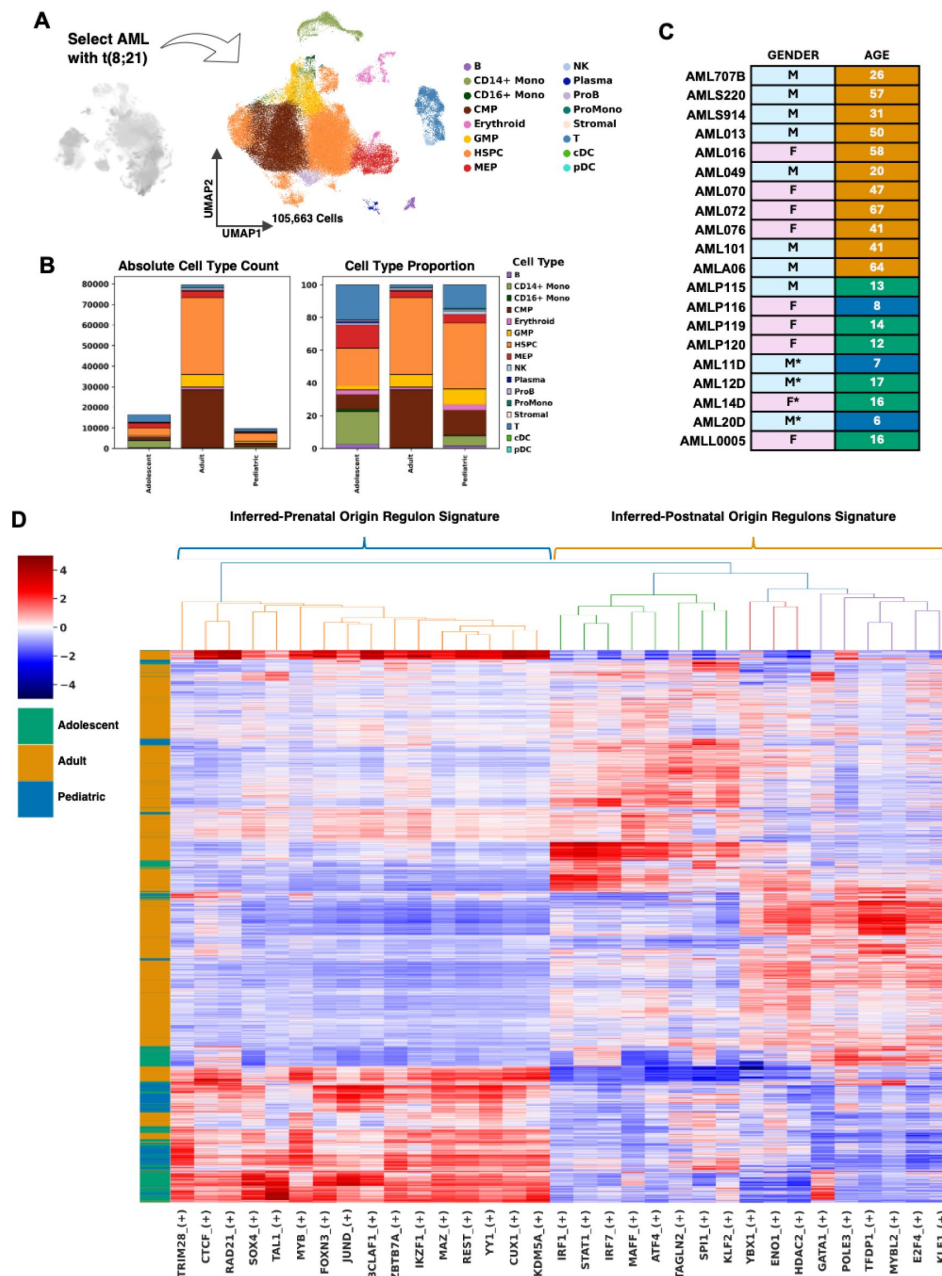


Figure 2 –

AML scAtlas Reveals Age-Associated Heterogeneity in t(8;21) AML

(A) Clinical characteristics of the AML scAtlas t(8;21) samples, which were divided into three age groups. Gender is indicated where available. *Sample sex inferred from XIST/ChrY expression. (B) Using the AML scAtlas t(8;21) sample cells, UMAP was re-computed and shows the different cell types. (C) Bar plots of the absolute cell type numbers (left panel) and the cell type proportions (right panel) stratified by age group. The CD34 enrichment performed on several adult samples is reflected. (D) Using HSPCs and CMPs only, the pySCENIC gene regulatory network (GRN) and regulon AUC scores were calculated. Z-score normalized scores underwent hierarchical clustering to create a clustered heatmap and identify age-associated regulons. Regulons were prioritized using their regulon specificity scores (RSS).

To define signature regulons, we used the clustered dendrogram to select the regulon clusters most associated with the pediatric (below 10 years-old) and adult samples (over 18 years-old) (Supplementary Figure 4C; [Figure 2D](#)). The pediatric regulon signature is proposed to represent *in-utero* origin t(8;21) AML (henceforth termed ‘inferred-prenatal’), and includes 16 regulons defined by a distinct group of hematopoietic transcription factors (TFs) (TRIM28, CTCF, RAD21, SOX4, TAL1, MYB, FOXN3, JUND, BCLAF1, ZBTB7A, IKZF1, MAZ, REST, YY1, CUX1, KDM5A), many of which have clearly defined roles in HSPCs and AML [\[45–47\]](#). The adult regulon signature, presumed representative of the postnatally acquired t(8;21) AML (henceforth termed ‘inferred-postnatal’), combines 3 discrete clusters of regulons (YBX1, ENO1, and HDAC2; GATA1, POLE3, TFDP1, MYBL2, E2F4, and KLF1; IRF1, STAT1, IRF7, MAFF, AFT4, TAGLN2, SPI1, and KLF2), defined by TFs previously implicated in various hematopoietic, leukemic and inflammatory processes [\[48–50\]](#). Importantly, both signatures contain key components of the AP-1 complex, which is heavily implicated in the biology of t(8;21) AML [\[51, 52\]](#) and undergoes dynamic changes during aging [\[53\]](#).

Samples of 6 individuals aged 12-17 clustered with the pediatric samples and showed enrichment for the inferred-prenatal signature ([Figure 2D](#)), which suggests that older adolescents (up to aged 17 in our cohort) more closely resemble pediatric AML with t(8;21) and remain biologically distinct from adult onset disease. This implies that the inferred in utero origin of t(8;21) AML can also be present in AML diagnosed in older children.

Validation of Age-Associated Regulons in Bulk-RNA-Seq Cohorts of t(8;21) AML

We next sought to perform external validation of our age-associated regulon signatures in a larger cohort of patients. We applied the AUCell algorithm from pySCENIC [\[22\]](#), to the t(8;21) AML bulk RNA-seq from the TARGET [\[2\]](#) and BeatAML [\[23\]](#) cohorts (n=103) to calculate the activity of our pediatric inferred-prenatal and adult inferred-postnatal regulons. Using unsupervised clustering of the bulk RNA-seq AUCell results, we identified discrete clusters of samples that were highly enriched for our inferred-prenatal and inferred-postnatal origin-associated regulons ([Figure 3A](#)).

Given the limitations of most scRNA-seq platforms in detecting lowly expressed genes, notably TFs, we used these bulk RNA-seq samples to refine our identified gene regulatory networks by detecting differentially expressed regulon-associated TFs. We used our inferred-prenatal and inferred-postnatal signature clusters and performed differential gene expression analysis between these samples, which are well separated by principal components analysis (PCA) ([Supplementary Figure 5A](#)). We then looked for differentially expressed regulon-associated TFs between the two groups; although changes in TF expression are subtle ([Supplementary Figure 5B](#); [Figure 3B](#)), we identify significantly differentially expressed TFs which reflect the observed differences in regulon activity and indicate the most critical regulons in our age-related GRN signatures ([Figure 3B](#)). This further delineated the inferred-prenatal signature to 5 key TFs (KDM5A, REST, BCLAF1, YY1, and RAD21), and the inferred-postnatal signature to 9 TFs (ENO1, TFDP1, MYBL2, KLF1, TAGLN2, KLF2, IRF7, SPI1, and YXB1).

We next performed gene set enrichment analysis (GSEA) using all significant differentially expressed genes, to investigate pathways which were enriched in the inferred-prenatal samples compared to the inferred-postnatal ones ([Figure 3C](#)). Notably, this identified an increase in stemness-associated genes, and in the SMARCA2 target genes, which is a key player in HSC gene expression regulation and chromatin remodeling [\[54\]](#). SMARCA2 is also known to be upregulated during the fetal-to-adult HSC transition [\[55\]](#), implying that the observed SMARCA2 enrichment may indeed reflect the inferred fetal HSC cell-of-origin. Genes impacted by YY1 depletion were also downregulated compared to the samples of inferred-postnatal leukemia origin, which supports the identification of YY1 as an inferred-prenatal regulon ([Figure 3C](#)). We also performed GSEA

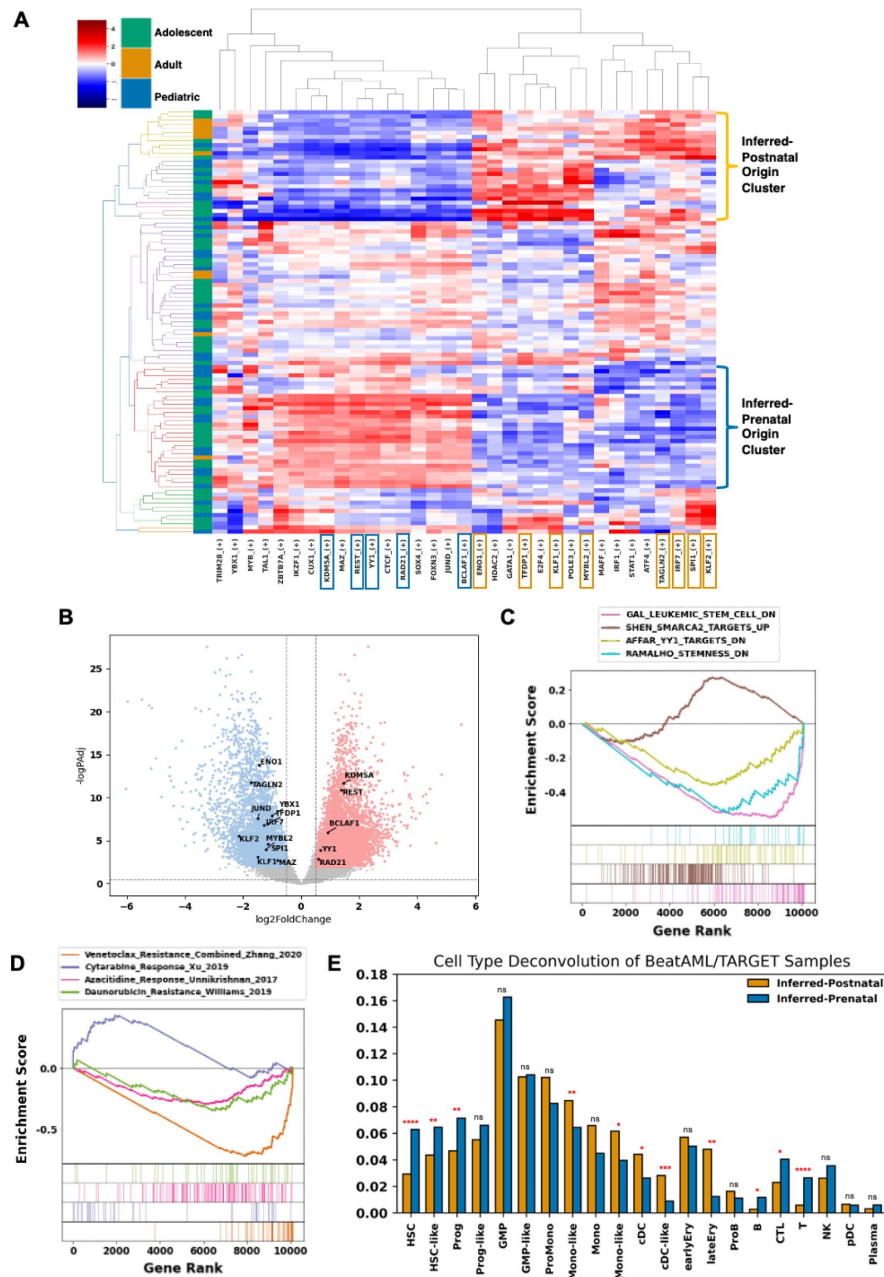


Figure 3 –

Validation of Age-Associated Regulons in Large Bulk RNA-Seq Cohorts

(A) Using previously defined age-associated regulons, pySCENIC AUC scores (Z-score normalized) were clustered to identify samples most enriched for inferred-prenatal and inferred-postnatal origin signatures. (B) Volcano plot of differentially expressed genes when comparing the inferred-prenatal origin and inferred-postnatal origin samples. Adjusted P value threshold 0.01; log₂ fold change threshold 0.5. Regulon signature associated TFs are indicated. (C) Enrichment plot of significant gene sets enriched in the inferred-prenatal origin samples. GSEA was performed on the DEGs using MSigDB databases. FDR q-value threshold <0.05. (D) Enrichment plot of drug sensitivity gene sets enriched in the inferred-prenatal origin samples. GSEA was performed on the DEGs, using drug response signatures from published studies of 4 widely used AML drugs. FDR q-value threshold <0.05. (E) The predicted cell type proportions estimated using AutoGeneS deconvolution, of the inferred-prenatal and inferred-postnatal origin samples were compared using T-Tests. Significant P values <0.05 (*), <0.01 (**), <0.001 (***) and <0.0001 (****) are indicated.

using drug response signatures from published studies [56–59], to deduce whether the inferred-prenatal or inferred-postnatal origin of the leukemia affects chemosensitivity related genes (**Figure 3D**). This analysis revealed that inferred-prenatal origin t(8;21) AML is enriched for genes associated with increased chemosensitivity to cytarabine, venetoclax, and daunorubicin (**Figure 3D**).

We hypothesized that the increase in stemness-associated genes in the leukemia with the inferred-prenatal origin could be reflective of potential differences in the leukemic cell-of-origin and its impact on myeloid differentiation. To compare the cellular heterogeneity between inferred-prenatal and inferred-postnatal samples, we performed cell type deconvolution using AutoGenes [60], with a curated LSPC reference profile [7], to compare the cellular heterogeneity between prenatal and postnatal origin bulk RNA-seq samples (**Figure 3E**). This revealed a higher proportion of most HSPC cell types (HSC, HSC-like, LSPC-Cycle, LSPC-Primed, LSPC-Quiescent), with a reduction in differentiated myeloid cell types (CMP, MEP, GMP, ProMono, CD14+ Mono, CD16+ Mono, cDC, Erythroid) in the leukemia data with inferred-prenatal origin (**Figure 3E**). To corroborate this finding, we compared cell type proportions in the original t(8;21) subset of AML scAtlas, confirming that cells with the inferred-prenatal signature comprise more HSC/HSC-like cells than inferred-postnatal signature cells (**Supplementary Figure 5C;5D**). However, comparison of cell type proportions in this dataset is confounded by differences in sample processing as some studies performed CD34 selection, hence there is more cell type diversity observed in the pediatric samples (**Supplementary Figure 5C;5D**).

Multomics Single-Cell Data Reveals a Denoised GRN and Identifies Candidate Perturbations in Prenatal Origin t(8;21) AML

We used the scRNA-seq and scATAC-seq data from a recent cohort of pediatric t(8;21) AML patients [24] at multiple clinical time points to uncover the eGRN (enhancer-driven GRN) in inferred-prenatal and inferred-postnatal origin t(8;21) AML. Initially, we identified the two most representative samples of our inferred-prenatal and inferred-postnatal signatures by using pySCENIC AUCell [22] to measure the activity of our previously defined regulons. Unsupervised clustering of the AUC scores was used to infer whether each sample matched the regulon signatures, identifying one inferred-prenatal sample and one inferred-postnatal sample for downstream analysis (**Supplementary Figure 6A**).

SCENIC+ [25] analysis uses both scRNA-seq and scATAC-seq to identify candidate enhancer regions and TF-binding motifs, which are then used to link TFs to target genes and identified enhancers. This creates enhancer-driven regulons (eRegulons), forming an eGRN. We applied SCENIC+ [25] to the leukemia samples with the inferred-prenatal and inferred-postnatal origin at diagnosis and relapse, keeping only regulons that showed a correlation between both modalities to retain only the most robust regulons (**Supplementary Figure 6B**). This revealed several eRegulons across both patients (**Supplementary Figure 6C**), many of which were patient specific, particularly when comparing HSPC populations (**Figure 4A**). The sample of inferred-prenatal origin shows a specific HSC eRegulon profile, whereas the inferred-postnatal sample more closely resembles the corresponding Granulocyte-Monocyte Progenitor (GMP) (**Figure 4A**). Interestingly, at relapse the inferred-prenatal origin patient undergoes a chemotherapy-driven lineage switch to a lymphoid phenotype, which may suggest that the leukemia originated from a less committed progenitor (**Figure 4A**).

To identify clusters of closely related eRegulons, we computed the correlations between eRegulons enrichment. We identified 5 main clusters of eRegulons which correspond to different inferred-signature samples (**Figure 4B**). For each eRegulon cluster, we used the associated target genes as input for gene ontology over representation analysis (ORA), to compare functional differences in the eGRN. This revealed fundamental differences in the underlying biological processes (**Figure 4C**; **Supplementary Figure 7A**). The AML sample with inferred-prenatal origin shows critical

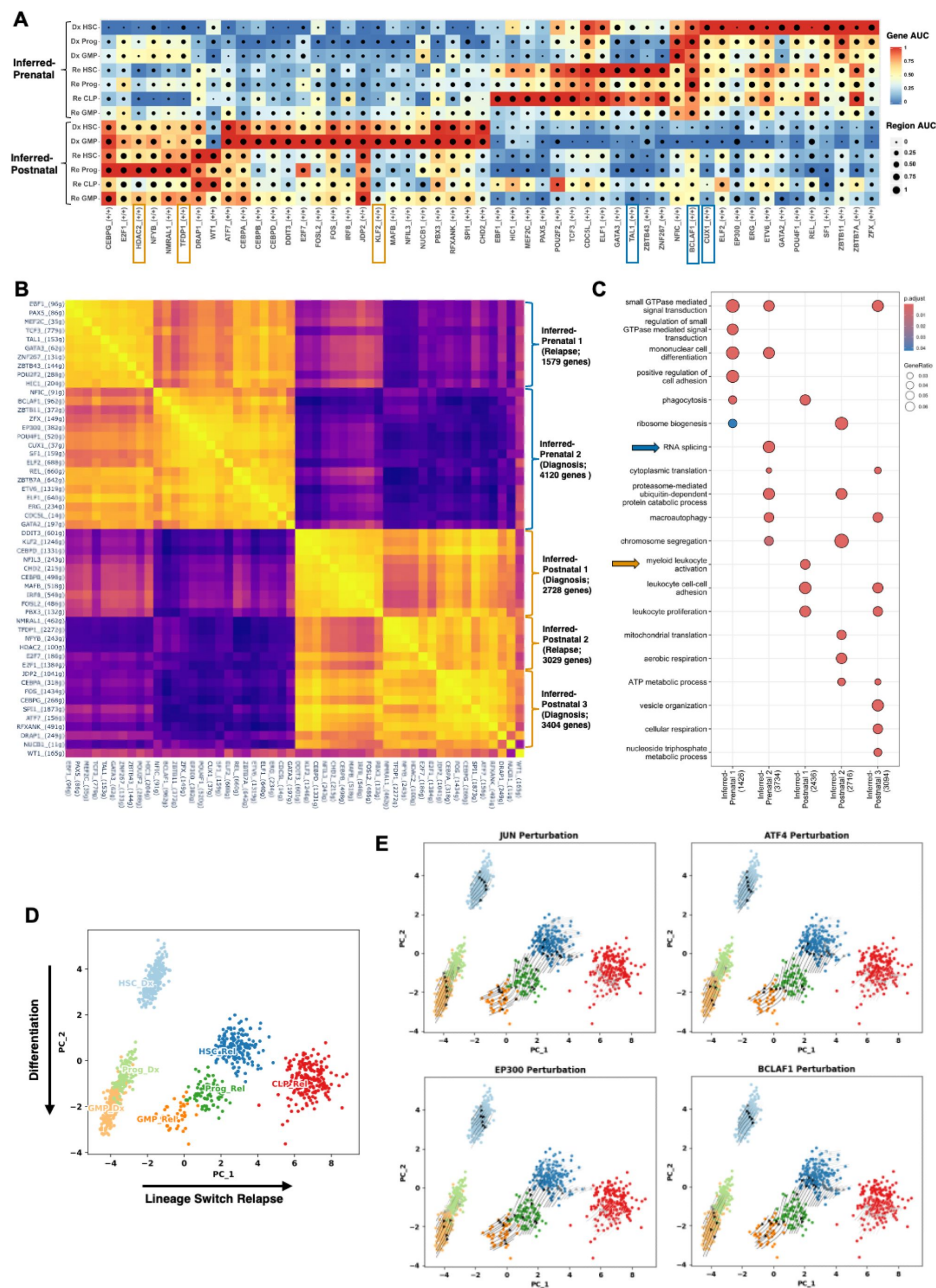


Figure 4 –

Combining Multiomics Data Interrogates Age-Associated Regulons

(A) SCENIC+ eRegulon dotplot of showing correlation between scRNA-seq target gene activity (indicated by the color scale) and scATAC-seq target region accessibility (depicted by spot size). RSS identified the key activating eRegulons (+/-) between inferred-prenatal and inferred-postnatal origin disease and allows comparison of diagnosis (Dx) and relapse (Rel) time points. (B) Correlation plot of the eRegulon target genes. Clusters reflecting inferred-prenatal and inferred-postnatal origin disease at diagnosis (Dx) and relapse (Rel) reflect co-binding of eRegulon TFs. (C) Over-representation analysis of age-associated eRegulon target genes using GO Biological Processes curated gene sets. Adjusted P value threshold 0.05. (D) Principal components analysis (PCA) of the gene based eRegulon enrichment scores for the inferred-prenatal origin disease at diagnosis and relapse. PC1 axis explains variance occurring between diagnosis and relapse, where this patient underwent a lineage switch. PC2 captures variance related to hematopoietic differentiation. (E) SCENIC+ perturbation simulation shows the predicted effect of knockout of selected TFs on the previously computed PCA embedding. Arrows indicate the predicted shift in cell states relative to the initial PCA embedding.

transcriptional components as the key molecular processes, such as RNA splicing, histone modification, and TF binding. Whereas AML sample with inferred-postnatal origin appears more immune-mediated, with leucocyte proliferation and leucocyte receptor activation (**Figure 4C**; **Supplementary Figure 7A**). This further supports the stemness signature in presumed prenatal origin t(8;21) AML, compared to postnatal origin disease.

Previous analysis using the TARGET^[23] and BeatAML^[23] datasets indicated that inferred-prenatal and inferred-postnatal origin t(8;21) AML may harbor different levels of chemosensitivity based on published drug response signatures (**Figure 3D**). Therefore, we performed in silico perturbations of eRegulon-associated TFs. PCA of the diagnosis and relapse samples recapitulates expected differentiation trajectories along PC2, while capturing differences between diagnosis and relapse along PC1 (**Figure 4D**). Using the SCENIC+^[25] perturbation simulation workflow to model the predicted effects of TF perturbation on cell state, we identified TFs estimated to induce differentiation, as defined by a negative shift in PC2 (**Supplementary Figure 7B**). We prioritized TFs predicted to induce an effect on the HSC compartment and identified 18 TFs predicted to induce a significant effect on HSC differentiation (**Supplementary Figure 7C**). Several of these TFs are components of the AP-1 complex (JUN, ATF4, FOSL2), which are established targets of the t(8;21) fusion protein and are known to propagate t(8;21) AML^[51, 61] (**Supplementary Figure 7C**; **Figure 4E**).

EP300 was one of the most impactful hits, which has recently had a role established in driving t(8;21) AML biology through acetylation permitting self-renewal^[62]. This suggests that presumed prenatal origin pediatric t(8;21) AML may be particularly sensitive to EP300 inhibition. One of the most striking predictions, for both diagnostic and relapse HSC populations, was BCLAF1 (**Supplementary Figure 7C**; **Figure 4E**). BCLAF1 is a regulator of normal HSPCs^[63], and its expression level declines during hematopoietic differentiation. While recent studies have identified a role for BCLAF1 in AML^[64], this has not been explored in detail in the context of pediatric AML or t(8;21) AML, and may present a therapeutic opportunity.

We also performed SCENIC+^[25] perturbation modelling on the postnatal origin sample (AML12). PCA was more difficult to interpret in this case, as branching differentiation trajectories towards a lymphoid or myeloid fate appear along the PC2 axis, while PC1 distinguishes diagnosis and relapse samples (**Supplementary Figure 7D**). Therefore, we prioritized TFs based on a predicted effect similar to AP-1 complex components, as it is known that this complex is a critical regulator in t(8;21) AML. We identified several TFs which were identified in our original postnatal origin signature as likely to have an effect (**Supplementary Figures 7D-F**), indicating our previous analyses identified biologically relevant GRNs.

To investigate the impact EP300 and BCLAF1 in more detail, we used the DepMap^[65] database to assess the dependency of t(8;21) AML cell lines to these genes (**Supplementary Figure 7G**). We found that the two widely-used cell lines of t(8;21) AML, KASUMI-1 (7-year-old donor)^[66] and SKNO-1 (22-year-old donor)^[67], were among the most sensitive to these perturbations based on their DepMap effect scores. Together this indicates that EP300 inhibition may be particularly effective in t(8;21) AML, and that BCLAF1 may present a new therapeutic target for t(8;21) AML, particularly in children with inferred pre-natal origin of the driver translocation.

Discussion

Here we have generated a new data resource, AML scAtlas, to investigate single-cell biology across a range of subtypes of AML. By including 222 samples comprising 748,679 cells of patients with a wide range of clinical characteristics, we overcome the limitations of many standalone single-cell studies by allowing AML subtype-focused analysis with enough data for robust statistical

comparisons. The AML scAtlas dataset is publicly available (<https://cellxgene.bmh.manchester.ac.uk/AML/>) to allow researchers from all areas of AML, to utilize this dataset to answer their own biological questions and generate new hypotheses.

To further address a clinically relevant question using this data source, we compare differences between pediatric and adult-onset disease based on the potential biological effect of the in-utero origin of pediatric leukemia. Data of our AML scAtlas was used to explore the GRNs in adult and pediatric t(8;21) AML, and revealed a strong age-associated GRN signature. This suggests that while pediatric and adult t(8;21) AML is propagated by the same driver translocation, there are clear biological differences which correlate with age. This may be due to differences in the cell-of-origin, which has been shown to transform from a HSC or a more lineage restricted GMP in t(8;21) AML mouse models [68]. As pediatric t(8;21) can arise in-utero, as evidenced by previous studies [17], and adult t(8;21) is acquired postnatally [18, 19], we propose that the observed age-related differences in AML with t(8;21) reflect these differences in the developmental origins of the disease. Here, we find two distinct groups of regulons which best reflect inferred-prenatal origin and inferred-postnatal origin disease. Our cohort is the largest scRNA-seq dataset to explore t(8;21) AML biology to date, however, the number of patients included remains low (n=22), and many of the studies containing the adult samples used CD34 selection in their experimental protocol creating a bias towards HSPCs in these samples.

To overcome some of these limitations, we used bulk RNA-seq samples from the TARGET [2] and BeatAML [23] studies with t(8;21) AML (n=103) to validate our regulon signatures. This identifies two clusters of samples which closely match these signatures, showing that the regulon patterns identified from our AML scAtlas are also applicable to bulk RNA-seq data which allows exploration of larger patient cohorts. We compared the transcriptomes of the groups with inferred-prenatal and inferred-postnatal origin, to prioritize TFs which were differentially expressed and to compare differences in underlying biology and drug response. We identified 5 signature TFs (KDM5A, REST, BCLAF1, YY1, RAD21) for inferred-prenatal origin disease, some of which have roles in embryonic stem cells [69, 70], and critical functions in the maintenance of HSCs [45–47]. Whereas TFs identified in inferred-postnatal origin samples, such as interferon regulatory factors (IRFs), HDAC2, and SPI1, reflect inflammatory and immune processes, many of which have been implicated in leukemia [48–50]. We also found that inferred-prenatal origin samples had a higher proportion of HSC/HSC-like cell types compared to inferred-postnatal origin samples, which suggests that prenatal origin t(8;21) AML confers a more primitive state compared to postnatal onset t(8;21) AML cases, supporting age-associated differences in the cell-of-origin playing a role in disease biology.

Given these biological differences, we used bulk RNA-seq to predict chemosensitivity using published drug response signatures [56–59], finding that the inferred-prenatal samples were enriched for genes indicative of cytarabine sensitivity, and depleted of genes suggestive of daunorubicin and venetoclax resistance. Our analysis further suggests that the developmental origins of the disease may be important for drug response signatures, which might inform the design of novel therapeutic strategies and provides further biological evidence that pediatric AML might benefit from different clinical management compared with adult-onset AML. Importantly, venetoclax is currently in the AML23 trial (NCT05955261); this data supports further evaluation of venetoclax treatment in pediatric t(8;21) AML.

Using an additional single-cell multiomic dataset, using SCENIC+ we reconstructed the enhancer-driven regulons (eRegulons) in samples matching our inferred-prenatal and inferred-postnatal regulon signatures. Upon comparing eRegulons for each patient at diagnosis and relapse, we identified clusters of highly correlated eRegulons which were defined by different biological processes. This indicated that the samples with inferred-prenatal origin are propagated by transcriptional dysregulation, whereas the samples of inferred-postnatal origin are largely driven by inflammatory signaling. We used SCENIC+ to model the predicted impact of TF perturbations

on our prenatal origin sample at diagnosis and relapse. Importantly, we identified several key components of the AP-1 complex, which is critical in t(8;21) AML biology and is also associated with dynamic age-related transcriptional changes [51–53].

Through our analysis, we identified EP300 as a candidate target, which has been shown to be critical for t(8;21) AML biology [62]. This has been shown to cause an effect in KASUMI-1 and SKNO1 cell lines. EP300 has been identified as a promising therapeutic target in AML with several molecules in development [71]; our data indicates potential specific therapeutic benefit in prenatal origin t(8;21) AML. One of the most impactful perturbation predictions for the HSC compartment at diagnosis and relapse was BCLAF1. This is consistent with previous evidence of its importance in HSCs [63, 72] and AML [64], but has not been studied specifically in the context of pediatric AML or t(8;21) AML previously. The DepMap data shows that KASUMI-1 is the most sensitive myeloid cell line to BCLAF1 perturbation, and our GRN analyses suggest it is particularly active in pediatric t(8;21) AML of inferred in-utero origin, thus this may represent an additional prognostic indicator.

Conclusions

Overall, we demonstrate that large-scale single-cell data integration provides a powerful tool to explore specific patient groups in detail, allowing for robust comparisons. We present the AML scAtlas as a data resource for researchers to answer their own biological questions. Applying our AML scAtlas to t(8;21) AML, we uncover age-associated gene regulatory networks which we propose relates to differences in the developmental origins, biology and outcome of the disease, and might point to novel candidate therapeutic targets which may be more effective in pediatric t(8;21) AML compared to adult-onset disease.

Methods

For the complete analysis code, including the conda environments used for analysis, see GitHub Repo (<https://github.com/jesswhitts/AML-scAtlas>).

Data Collection

A literature search was performed for published AML scRNA-seq datasets [4–6, 8, 26–40]. Suitable studies were selected based on the data quality (over 1000 counts and 500 genes detected per cell for most of the data). Diagnostic, primary AML samples were selected from each AML study. Where healthy donor samples were present, these were also included, along with an additional 4 studies with healthy bone marrow samples.

Initial Data Processing

Each scRNA-seq dataset underwent initial quality control individually using Scanpy (v1.9.3) [73] as some studies provided raw data and others provided pre-filtered data. Where raw data was provided, doublets were removed using Scrublet (v0.2.3) [74] and cells were filtered using the median absolute deviation as described in this single-cell best practices handbook [10, 75].

Once filtered, datasets were combined, and quality control was performed using Scanpy (v1.9.3) [73]. The full dataset had quality thresholds applied (percentage mitochondrial counts <10, read counts >1000, gene counts >500), removing any samples which had fewer than 50 cells remaining after filtering. Genes present in <50 cells were removed. MATA1 was removed as this was highly abundant in many cells and considered artefactual.

Batch Correction

The presence of batch effects was determined through dimensionality reduction and clustering using Scanpy (v1.9.3) [73], and using the kBET algorithm (v0.99.6) [76]. This was repeated on individual studies, to assess whether there were sample-wise batch effects. Batch correction benchmarking was implemented using Harmony (Scanpy v1.9.3 implementation) [11], scVI (v1.0.3) [12], and scANVI (v1.0.3) [13] and quantified using scIB (1.1.4) [9]. Different numbers of highly variable genes were used to select the optimal number for integration. Batch correction was performed using scVI (v1.0.3) [12] with the top 2000 highly variable genes, using sample as the model covariate.

AML scAtlas Annotation

The scVI corrected embedding was used to run UMAP and Leiden clustering using Scanpy functions (v1.9.3) [73]. Cell type annotation was performed using CellTypist (v1.6.0) [77], SingleR (v2.0.0) [78], and scType (v1.0) [79], taking the consensus of the different tools and the expression of key marker genes to manually curate the cell types. To annotate LSCs, a custom SingleR (v2.0.0) [78] reference was created using the Zeng et al [7] revised annotations of the Van Galen et al [5] dataset. This was also correlated with the LSC6 [43] and LSC17 [44] scores for each cell.

AML with t(8;21) Analysis

Samples with the t(8;21) translocation were selected from the full AML scAtlas. The UMAP was re-computed, and genes were filtered to remove those detected in fewer than 50 cells for the revised dataset, leaving 24,866 genes remaining. Gene regulatory network analysis was performed using pySCENIC [21, 22] (v0.12.1) as per the recommended workflow. To facilitate comparisons between age groups, cell types were focused on HSPCs, as many adult samples were originally enriched for CD34. The RSS was calculated for the adult and pediatric samples to select the top 20 differential regulons per age group. Using SciPy hierarchical clustering (v1.12.0), regulons were filtered to identify regulon signature groups used for downstream analysis.

Bulk RNA-Seq Analysis

Bulk RNA-Seq data was downloaded for the TARGET [2] and BeatAML [23] cohorts and samples with t(8;21) were selected. Using the previously defined signature regulons, the AUCell algorithm [21] (v0.12.1) was implemented to measure regulon activity. Hierarchical clustering was performed using SciPy v1.12.0 to identify samples most enriched for each age-related signature. Differential gene expression analysis was implemented using DESeq2 [80] (v1.40.2), using a log2 fold change threshold of 0.5 and an adjusted p value cutoff of 0.01. Candidate differential genes, ranked on log2 fold change, underwent GSEA with GSEAPy (v0.10.8) using a significance threshold of 0.05. Cell type deconvolution was performed using AutoGeneS [60] (v1.0.4) using the recommended workflow. Significance when comparing groups was ascertained using a student's T-test on the predicted cell type proportion values for each sample.

Single-Cell Multi-Omics Analysis

The Lambo et al [24] scRNA-seq and scATAC-seq data was downloaded, and the t(8;21) samples selected. Using our previously defined signature regulons, the AUCell algorithm [21] (v0.12.1) was implemented to measure regulon activity. This identified the samples most enriched for each age-related signature as AML16 and AML12.

The SCENIC+ pipeline [25] (v1.0a1) was implemented as per the recommended Snakemake workflow for creating pseudo-multiome data. Regulons were filtered by correlation between modalities, using a threshold of 0.2 for non-multiome data. The most robust regulons were

prioritized based on the SCENIC+ recommendations (direct +/-) [25]. To facilitate comparisons, the eRegulon RSS was calculated for each patient and the top 30 eRegulons selected. The correlation between the gene sets underpinning eRegulons was calculated and sample-associated clusters were selected for over-representation analysis with clusterProfiler [81] (v4.8.3).

To predict the impact of specific TF perturbations on key cell types, SCENIC+ [25] perturbation modelling was implemented using the recommended parameters. TFs were then prioritized on their predicted impact on HSC differentiation and visualised using the PCA embedding. Candidate targets EP300 and BCLAF1 were queried in the DepMap [65] databases to infer their potential importance.

Availability of Data and Materials

The AML scAtlas is hosted online for public use (<https://cellxgene.bmh.manchester.ac.uk/AML/>). The processed AnnData object is also available to download from figshare (DOI: 10.48420/27269946). Details of all data used in this study can be found in Supplementary Table 1, along with associated links to the original data. All code used to perform the analyses presented here can be accessed in the GitHub repository: (<https://github.com/jesswhitts/AML-scAtlas>).

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Additional information

Author Contributions

MI, SMB, and GL conceived the study and oversaw the research. SM guided sub-analyses and assisted with interpretation. JW did the data collection and implemented downstream analyses. JW wrote the first draft of the manuscript. All authors interpreted the data and edited the manuscript. All authors approved the final manuscript.

Additional files

Supplementary figures and tables [↗](#)

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Reviewer #1 (Public review):

Summary:

In this manuscript, the authors performed an integration of 48 scRNA-seq public datasets and created a single-cell transcriptomic atlas for AML (222 samples comprising 748,679 cells). This is important since most AML scRNA-seq studies suffer from small sample size coupled with high heterogeneity. They used this atlas to further dissect AML with t(8;21) (AML-ETO/RUNX1-RUNX1T1), which is one of the most frequent AML subtypes in young people. In particular, they were able to predict Gene Regulatory Networks in this AML subtype using pySCENIC, which identified the paediatric regulon defined by a distinct group of hematopoietic transcription factors (TFs) and the adult regulon for t(8;21). They further validated this in bulk RNA-seq with AUCell algorithm and inferred prenatal signature to 5 key TFs (KDM5A, REST, BCLAF1, YY1, and RAD21), and the postnatal signature to 9 TFs (ENO1, TFDP1, MYBL2, KLF1, TAGLN2, KLF2, IRF7, SPI1, and YXB1). They also used SCENIC+ to identify enhancer-driven regulons (eRegulons), forming an eGRN, and found that prenatal origin shows a specific HSC eRegulon profile, while a postnatal origin shows a GMP profile. They also did an in silico perturbation and found AP-1 complex (JUN, ATF4, FOSL2), P300, and BCLAF1 as important TFs to induce differentiation. Overall, I found this study very important in creating a comprehensive resource for AML research.

Strengths:

(1) The generation of an AML atlas integrating multiple datasets with almost 750K cells will further support the community working on AML.

(2) Characterisation of t(8;21) AML proposes new interesting leads.

Weaknesses:

Were these t(8;21) TFs/regulons identified from any of the single datasets? For example, if the authors apply pySCENIC to any dataset, would they find the same TFs, or is it the increase in the number of cells that allows identification of these?

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Reviewer #2 (Public review):

Summary:

The authors assemble 222 publicly available bone marrow single-cell RNA sequencing samples from healthy donors and primary AML, including pediatric, adolescent, and adult patients at diagnosis. Focusing on one specific subtype, t(8;21), which, despite affecting all age classes, is associated with better prognosis and drug response for younger patients, the authors investigate if this difference is reflected also in the transcriptomic signal. Specifically, they hypothesize that the pediatric and part of the young population acquires leukemic mutations in utero, which leads to a different leukemogenic transformation and ultimately to differently regulated leukemic stem cells with respect to the adult counterpart. The analysis in this work heavily relies on regulatory network inference and clustering (via SCENIC tools), which identifies regulatory modules believed to distinguish the pre-, respectively, post-natal leukemic transformation. Bulk RNA-seq and scATAC-seq datasets displaying the same signatures are subsequently used for extending the pool of putative signature-specific TFs and enhancer elements. Through gene set enrichment, ontology, and perturbation simulation, the authors aim to interpret the regulatory signatures and translate them into potential onset-specific therapeutic targets. The putative pre-natal signature is associated with increased chemosensitivity, RNA splicing, histone modification, stem-ness marker SMARCA2, and potentially maintained by EP300 and BCLAF1.

Strengths:

The main strength of this work is the compilation of a pediatric AML atlas using the efficient Cellxgene interface. Also, the idea of identifying markers for different disease onsets, interpreting them from a developmental angle, and connecting this to the different therapy and relapse observations, is interesting. The results obtained, the set of putative up-regulated TFs, are biologically coherent with the mechanisms and the conclusions drawn. I also appreciate that the analysis code was made available and is well documented.

Weaknesses:

There were fundamental flaws in how methods and samples were applied, a general lack of critical examination of both the results and the appropriateness of the methods for the data at hand, and in how results were presented. In particular:

(1) Cell type annotation:

a) The 2-phase cell type annotation process employed for the scRNA-seq sample collection raised concerns. Initially annotated cells are re-labeled after a second round with the same cell types from the initial label pool (Figure 1E). The automatic annotation tools were used

without specifying the database and tissue atlases used as a reference, and no information was shown regarding the consensus across these tools.

b) Expression of the CD34 marker is only reported as a selection method for HSPCs, which is not in line with common practice. The use of only is admitted as a surface marker, while robust annotation of HSPCs should be done on the basis of expression of gene sets.

c) During several analyses, the cell types used were either not well defined or contradictory, such as in Figure 2D, where it is not clear if pySCENIC and AUC scores were computed on HSPCs alone or merged with CMPs. In other cases, different cell type populations are compared and used interchangeably: comparing the HSPC-derived regulons with bulk (probably not enriched for CD34+ cells) RNA samples could be an issue if there are no valid assumptions on the cell composition of the bulk sample.

(2) Method selection:

a) The authors should explain why they use pySCENIC and not any other approach. They should briefly explain how pySCENIC works and what they get out in the main text. In addition they should explain the AUCell algorithm and motivate its usage.

b) The obtained GRN signatures were not critically challenged on an external dataset. Therefore, the evidence that supports these signatures to be reliable and significant to the investigated setting is weak.

(3) There are some issues with the analysis & visualization of the data.

(4) Discussion:

a) What exactly is the 'regulon signature' that the authors infer? How can it be useful for insights into disease mechanisms?

b) The authors write 'Together this indicates that EP300 inhibition may be particularly effective in t(8;21) AML, and that BCLAF1 may present a new therapeutic target for t(8;21) AML, particularly in children with inferred pre-natal origin of the driver translocation.' I am missing a critical discussion of what is needed to further test the two targets. Put differently: Would the authors take the risk of a clinical study given the evidence from their analysis?

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Author response:

Reviewer #1 (Public review):

Summary:

In this manuscript, the authors performed an integration of 48 scRNA-seq public datasets and created a single-cell transcriptomic atlas for AML (222 samples comprising 748,679 cells). This is important since most AML scRNA-seq studies suffer from small sample size coupled with high heterogeneity. They used this atlas to further dissect AML with t(8;21) (AML-ETO/RUNX1-RUNX1T1), which is one of the most frequent AML subtypes in young people. In particular, they were able to predict Gene Regulatory Networks in this AML subtype using pySCENIC, which identified the paediatric regulon defined by a distinct group of hematopoietic transcription factors (TFs) and the adult regulon for t(8;21). They further validated this in bulk RNA-seq with AUCell algorithm and inferred prenatal signature to 5 key TFs (KDM5A, REST, BCLAF1, YY1, and RAD21), and the postnatal signature to 9 TFs (ENO1, TFDP1, MYBL2, KLF1, TAGLN2, KLF2, IRF7, SPI1, and YXB1). They also used SCENIC+ to identify enhancer-driven regulons (eRegulons), forming an eGRN,

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Strengths:

(1) The generation of an AML atlas integrating multiple datasets with almost 750K cells will further support the community working on AML.

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We thank the reviewer for a succinct summary of our work and highlighting its strengths.

Weaknesses:

Were these t(8;21) TFs/regulons identified from any of the single datasets? For example, if the authors apply pySCENIC to any dataset, would they find the same TFs, or is it the increase in the number of cells that allows identification of these?

The purpose of our study was to gain biological insights by integrating multiple datasets, to overcome limitations from small sample size. We expect that the larger dataset would improve network inference, which is what we implemented in the manuscript, hence we have not looked at individual datasets. However, we will investigate this further in the revised manuscript by running pySCENIC on individual datasets and comparing to the results drawn from the whole atlas.

Reviewer #2 (Public review):

Summary:

The authors assemble 222 publicly available bone marrow single-cell RNA sequencing samples from healthy donors and primary AML, including pediatric, adolescent, and adult patients at diagnosis. Focusing on one specific subtype, t(8;21), which, despite affecting all age classes, is associated with better prognosis and drug response for younger patients, the authors investigate if this difference is reflected also in the transcriptomic signal. Specifically, they hypothesize that the pediatric and part of the young population acquires leukemic mutations *in utero*, which leads to a different leukemogenic transformation and ultimately to differently regulated leukemic stem cells with respect to the adult counterpart. The analysis in this work heavily relies on regulatory network inference and clustering (via SCENIC tools), which identifies regulatory modules believed to distinguish the pre-, respectively, post-natal leukemic transformation. Bulk RNA-seq and scATAC-seq datasets displaying the same signatures are subsequently used for extending the pool of putative signature-specific TFs and enhancer elements. Through gene set enrichment, ontology, and perturbation simulation, the authors aim to interpret the regulatory signatures and translate them into potential onset-specific therapeutic targets. The putative pre-natal signature is associated with increased chemosensitivity, RNA splicing, histone modification, stem-ness marker SMARCA2, and potentially maintained by EP300 and BCLAF1.

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The main strength of this work is the compilation of a pediatric AML atlas using the efficient Cellxgene interface. Also, the idea of identifying markers for different disease onsets, interpreting them from a developmental angle, and connecting this to the different therapy and relapse observations, is interesting. The results obtained, the set of

putative up-regulated TFs, are biologically coherent with the mechanisms and the conclusions drawn. I also appreciate that the analysis code was made available and is well documented.

We thank the reviewer for reviewing our work, and highlighting its key features, including creation of AML atlas, downstream analysis and interpretation for t(8;21) subtype.

We also appreciate useful critique of our paper provided below.

Weaknesses:

There were fundamental flaws in how methods and samples were applied, a general lack of critical examination of both the results and the appropriateness of the methods for the data at hand, and in how results were presented. In particular:

(1) Cell type annotation:

a) The 2-phase cell type annotation process employed for the scRNA-seq sample collection raised concerns. Initially annotated cells are re-labeled after a second round with the same cell types from the initial label pool (Figure 1E). The automatic annotation tools were used without specifying the database and tissue atlases used as a reference, and no information was shown regarding the consensus across these tools.

We believe that most of the reviewer's criticisms stem from a misunderstanding, and we apologize for not explaining certain aspects of our work more clearly.

The two types of cell type annotation applied were different and served distinct purposes:

- One was using general bone marrow/blood reference datasets to annotate blood subtype lineage clusters.
- The other was using a CD34 purified AML specific reference dataset which included leukaemia-associated annotations, to identify HSPC subpopulations. We also implemented this on a single-cell level to allow more robust identification of these rare populations in a large dataset.

This is probably not well explained in the methods and figure presentation. We will clearly indicate in the revised manuscript that different HSPC annotations represent separate analysis and will update the figures to highlight this. We will provide a comprehensive review of the annotation strategies implemented, including the automated tool outputs, which may be useful for the single-cell community.

b) Expression of the CD34 marker is only reported as a selection method for HSPCs, which is not in line with common practice. The use of only is admitted as a surface marker, while robust annotation of HSPCs should be done on the basis of expression of gene sets.

We used CD34 expression in conjunction with other cell type annotations and marker sets to identify LSCs, although results are same when we use HSPC annotated cells without condition on CD34 expression. In the revised manuscript, we will simplify this analysis to use HSPC clusters as suggested by the reviewer.

c) During several analyses, the cell types used were either not well defined or contradictory, such as in Figure 2D, where it is not clear if pySCENIC and AUC scores were computed on HSPCs alone or merged with CMPs. In other cases, different cell type populations are compared and used interchangeably: comparing the HSPC-derived regulons with bulk (probably not enriched for CD34+ cells) RNA samples could be an issue if there are no valid assumptions on the cell composition of the bulk sample.

As mentioned in the Methods, we only excluded lymphoid cell types from the pySCENIC analysis to overcome the bias that some samples were enriched using CD34 selection when preparing them for scRNA-seq. We will make this clearer in the text and figures of the revised manuscript. It is difficult to overcome this bias when using bulk RNA samples, which may explain why some of our samples do not fit into our defined signature groups. However, as we do not have access to primary samples ourselves, we cannot provide a better matched experimental cohort for validation.

(2) Method selection:

a) The authors should explain why they use pySCENIC and not any other approach. They should briefly explain how pySCENIC works and what they get out in the main text. In addition they should explain the AUCell algorithm and motivate its usage.

pySCENIC is state-of-the-art method for network inference from scRNA data and is widely used within the single-cell community (over 5000 citations for both versions of the SCENIC pipeline). The pipeline has been benchmarked as one of the top performers for GRN analysis (Nguyen et al, 2021. Briefings in Bioinformatics). AUCELL is a module within the pySCENIC pipeline to summarise the activity of a set of genes (a regulon) into a single number which helps compare and visualise different regulons. We agree with reviewer that this could have been more clearly explained within the manuscript. We will update text in the revised manuscript to add more explanation.

b) The obtained GRN signatures were not critically challenged on an external dataset. Therefore, the evidence that supports these signatures to be reliable and significant to the investigated setting is weak.

These signatures were inferred from the best suitable AML single-cell RNA datasets available to date, and we used two independent datasets to validate our findings (the TARGET AML bulk RNA sequencing cohort, and the Lambo et al. scRNA-seq dataset). To our knowledge, there are no other better suited datasets for validation. Experimental validations on patient samples are beyond the scope of this study.

(3) There are some issues with the analysis & visualization of the data.

We will provide new statistical tests to improve robustness of the analysis as well as presentation and visualization of the data in the revised manuscript.

(4) Discussion:

a) What exactly is the 'regulon signature' that the authors infer? How can it be useful for insights into disease mechanisms?

The 'regulon signature' here refers to a gene regulatory program (multiple gene modules, each defined by a transcription factor and its targets) which are specific to different age groups. Further investigation into this can be useful for understanding why patients of different ages confer a different clinical course. We will add more text on the utility of our discovered 'regulon signature' in the discussion section of revised manuscript.

b) The authors write 'Together this indicates that EP300 inhibition may be particularly effective in t(8;21) AML, and that BCLAF1 may present a new therapeutic target for t(8;21) AML, particularly in children with inferred pre-natal origin of the driver translocation.' I am missing a critical discussion of what is needed to further test the two targets. Put

differently: Would the authors take the risk of a clinical study given the evidence from their analysis?

Of course, many extensive studies would be required before these findings are clinically translatable. We can include some perspectives on what further work is required in terms of further experimental validation and potential subsequent clinical study.

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